

EE 660 VLSI Design Laboratory

Course Project

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1. Design a two-stage operational amplifier, the circuit diagram of which is shown above using UMC 65 models. You have to assume that all the transistors have a channel length of 60nm. The specifications of the design are given below.

Voltage Gain ≥ 15 dB

$V_{DD} = 1.1$ V

$f_{3dB} \geq 1$ MHz

$C_L = 200$ fF

Slew Rate ≥ 100 V/ μ s

$V_{out} \in [0.3, 0.9]$ V

ICMR $\in [0.6, 1.0]$ V

$P_{diss} \leq 0.5$ mW

Phase Margin $\geq 60^\circ$

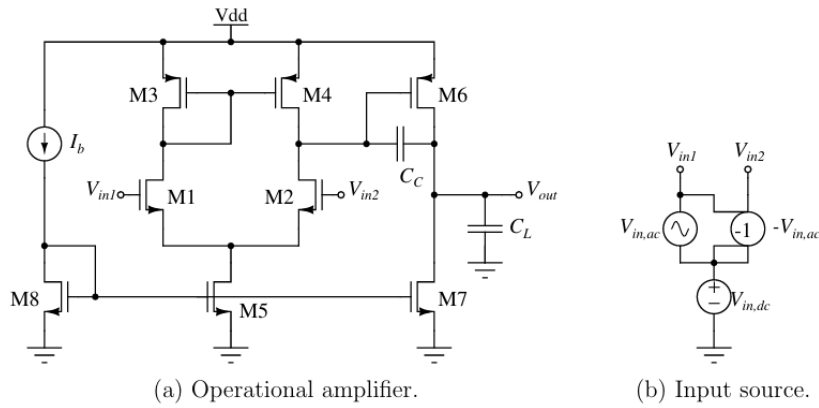
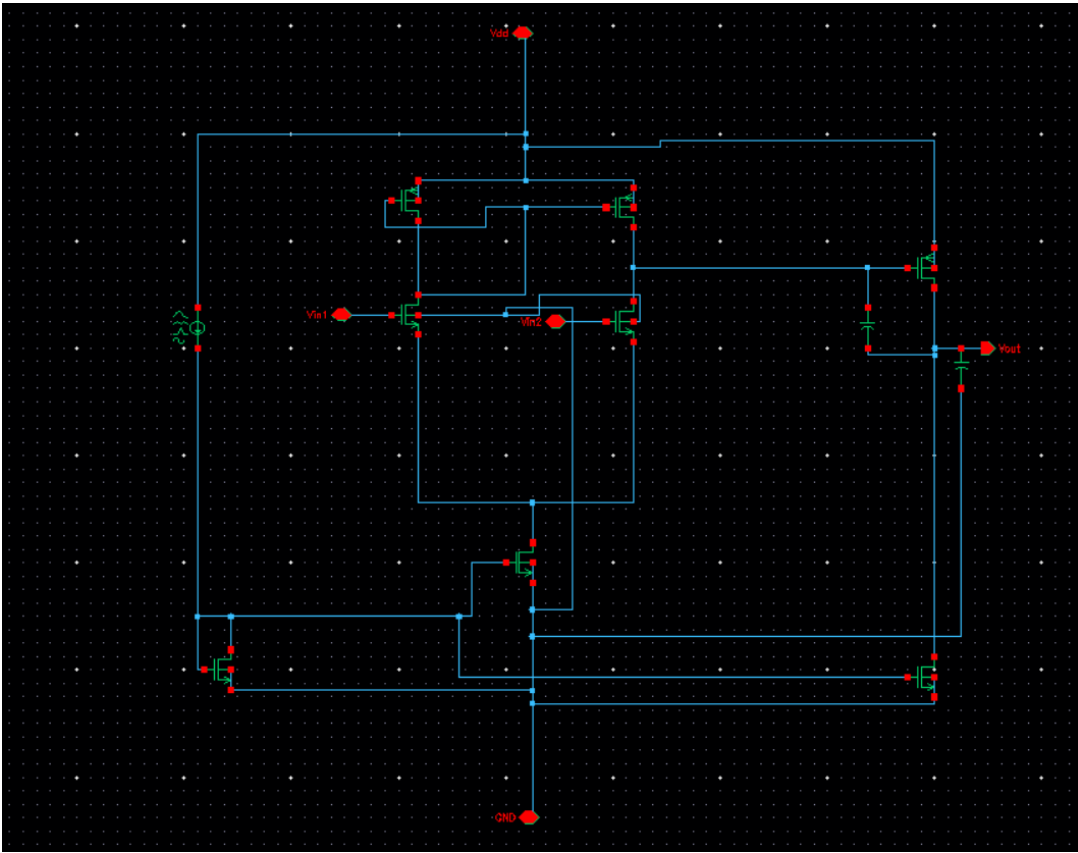
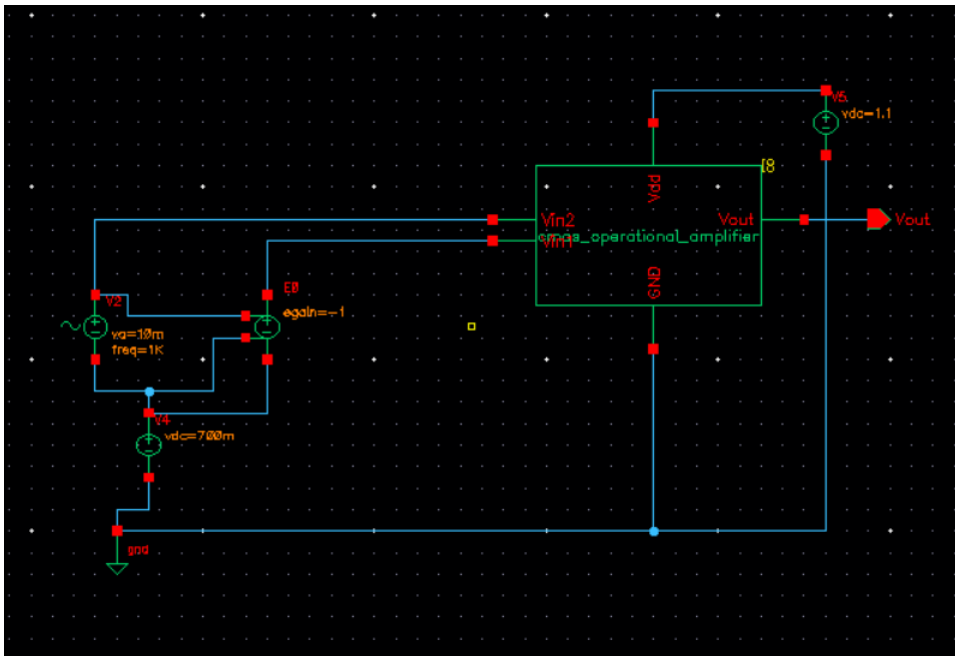


Figure 1: Schamatic



(a) Schematic in cadence



(b) Circuit

Figure 2: Schamatic

2. You will have to perform operating point analysis, DC analysis, AC analysis, and transient analysis to verify the functionality of the design. Your report is expected to have the results of all these simulations

Ans: Overview: We designed and validated the circuit through various analyses: DC, AC, transient, and operating point. The design was iteratively tuned to meet all specifications.

Operating Point Analysis: All transistors were confirmed to be in saturation. Refer to Figures 3 to 10.

ICMR Analysis: Figure 11 shows the output vs. input sweep and its derivative. Based on gain response, the ICMR is approximately 560mV to 870mV.

AC Analysis: From the gain and phase plot, the 3dB bandwidth is 38.52 MHz, and the phase margin is computed using phase at this 3 dB frequency:

$$PM = 180^\circ + (-32.8547^\circ) = 147.15^\circ$$

Refer to Figure 13.

Transient Analysis (Slew Rate): Slew rates were found to be +630.757 V/ μ s and -118.333 V/ μ s, shown in Figure 14.

Output Swing: The output voltage swing was measured as 312.6 mV to 509.3 mV, meeting the specified range. (Figure 15)

Power Dissipation: Calculated as $P = V_{DD} \times I_{DD} = 1.1 \text{ V} \times 419.47 \mu\text{A} = 461.4 \mu\text{W}$. (Figure 12)

Final Device Parameters:

Parameter	Value	Unit
W1, W2	1.8	μm
W3, W4	1.0	μm
W5	2.8	μm
W6	1.8	μm
W7	2.6	μm
W8	3.4	μm
C_c	120	fF
I_{ref}	200	μA
V_{DD}	1.1	V

Performance Summary:

- Voltage Gain: 27.08 dB
- f_{3dB} : 38.52 MHz
- Slew Rate: +630.757 V/ μ s, -118.333 V/ μ s
- Output Swing: 312.6 mV to 509.3 mV
- ICMR: 560 mV to 870 mV
- Phase Margin: 147.15°
- Power Dissipation: 461.4 μ W

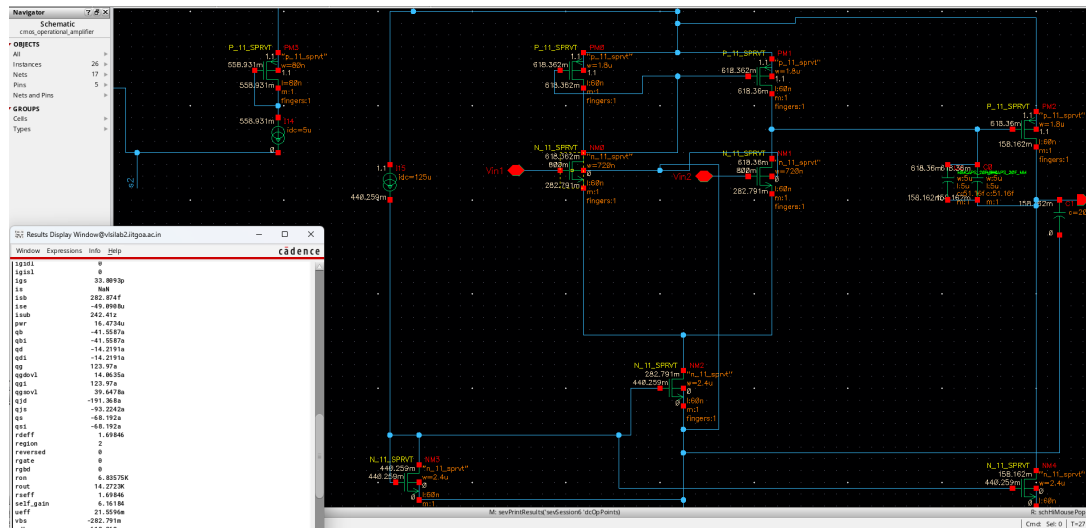


Figure 3: Saturation check for M1

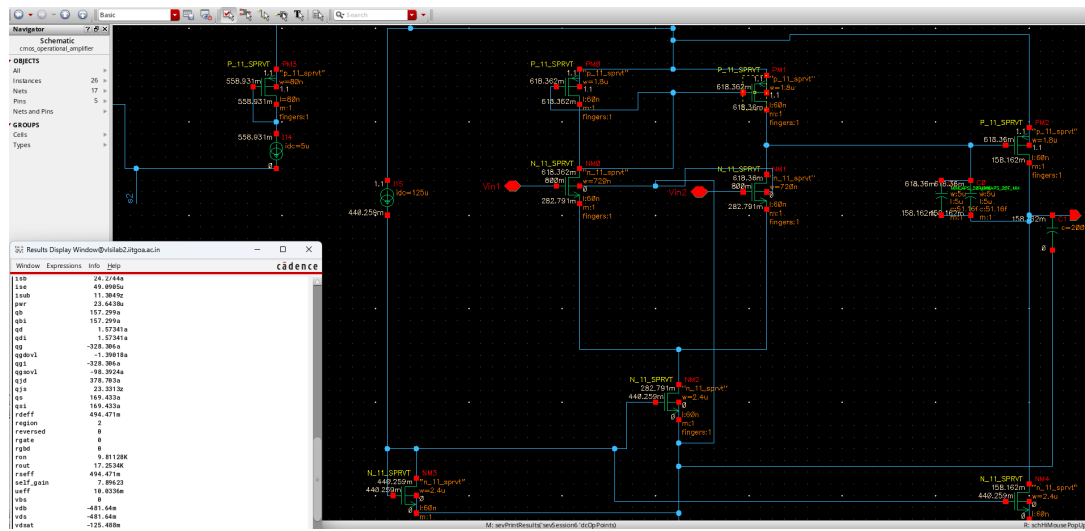


Figure 6: Saturation check for M4

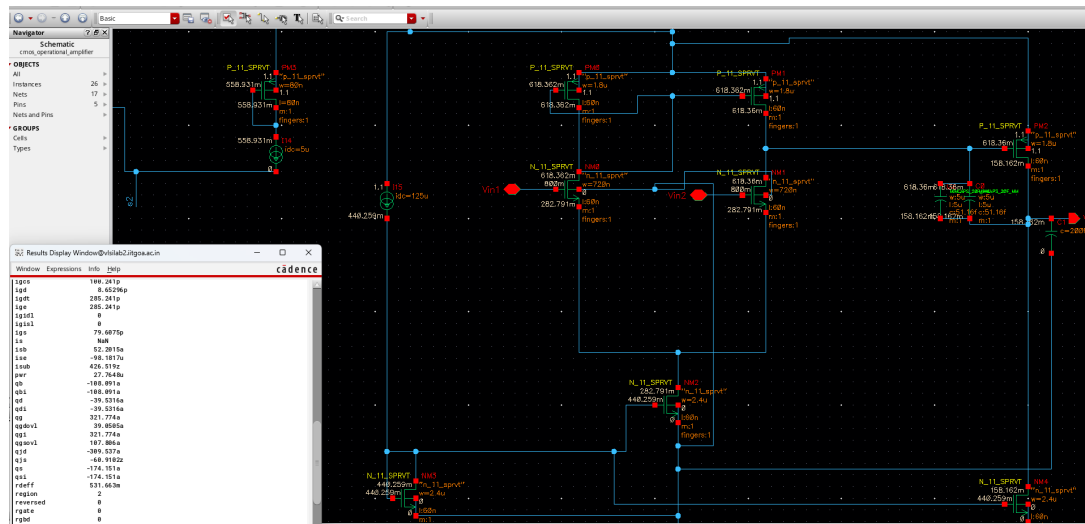
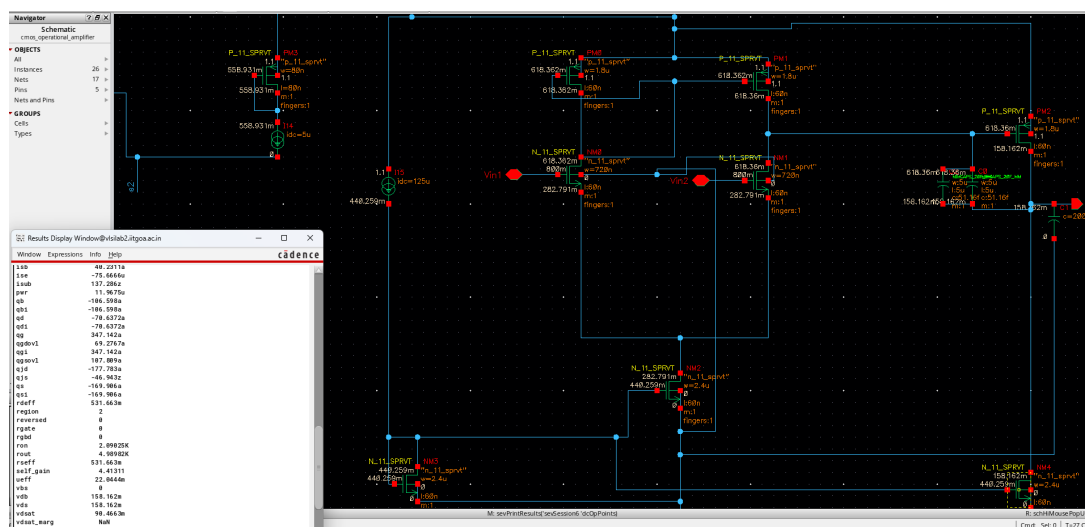
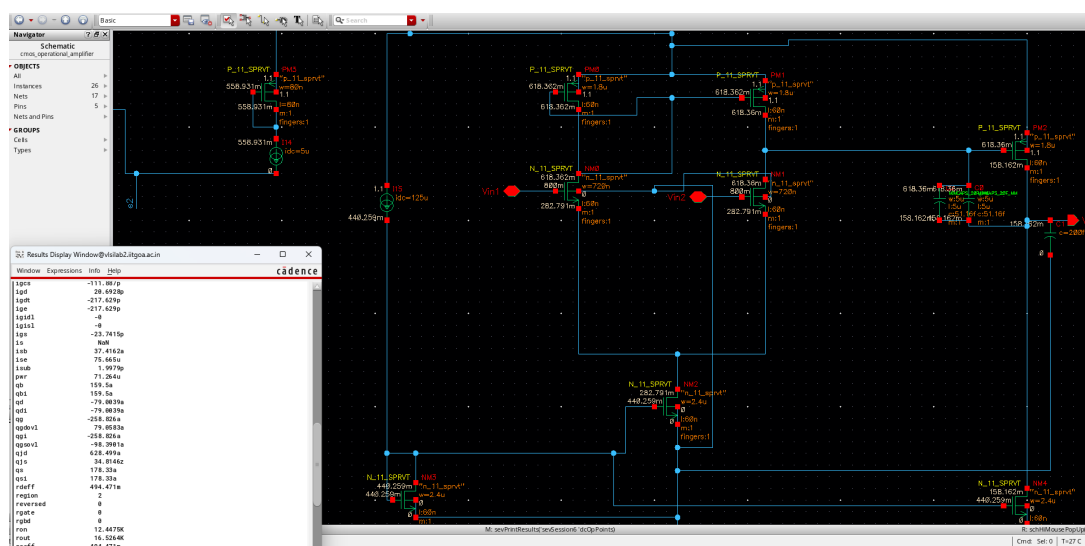


Figure 7: Saturation check for M5



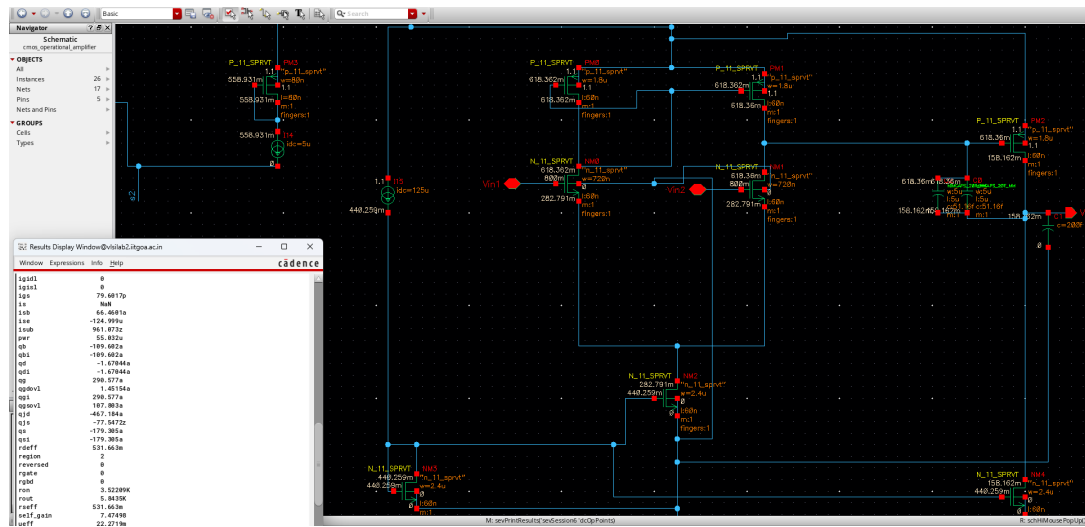


Figure 10: Saturation check for M8

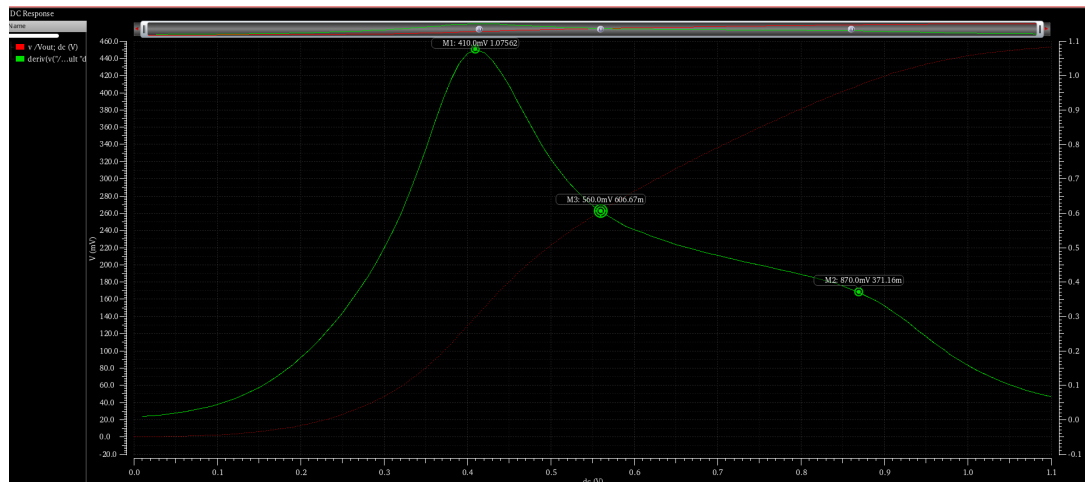


Figure 11: ICMR plot showing range from 560mV to 870mV

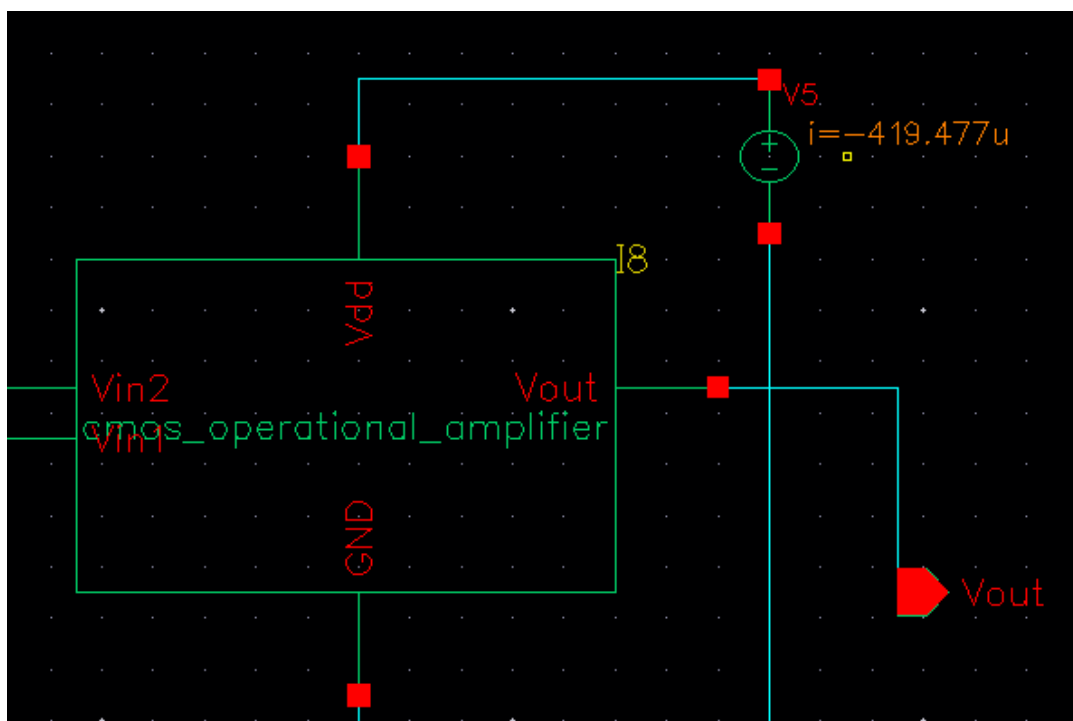


Figure 12: Power dissipation calculation from Cadence simulation

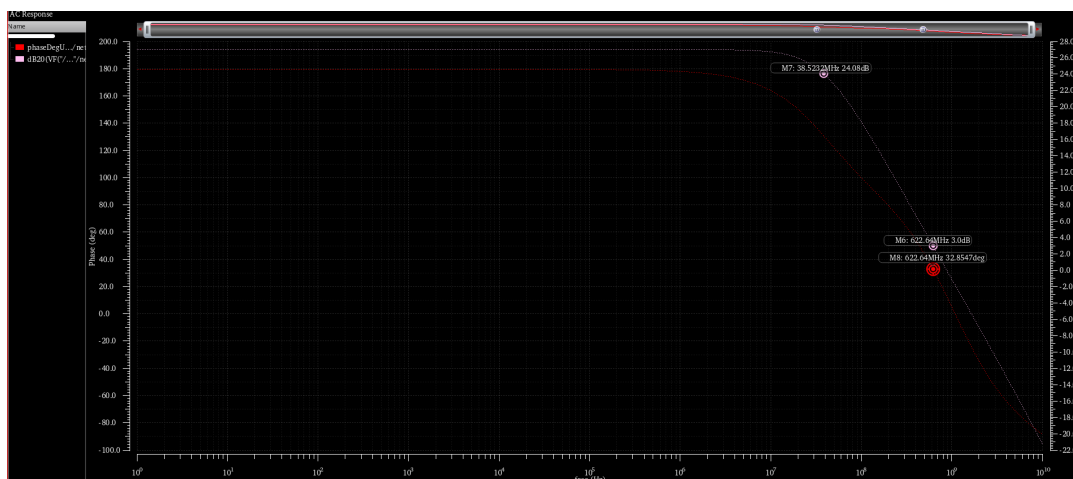


Figure 13: AC frequency response used to determine bandwidth and phase margin

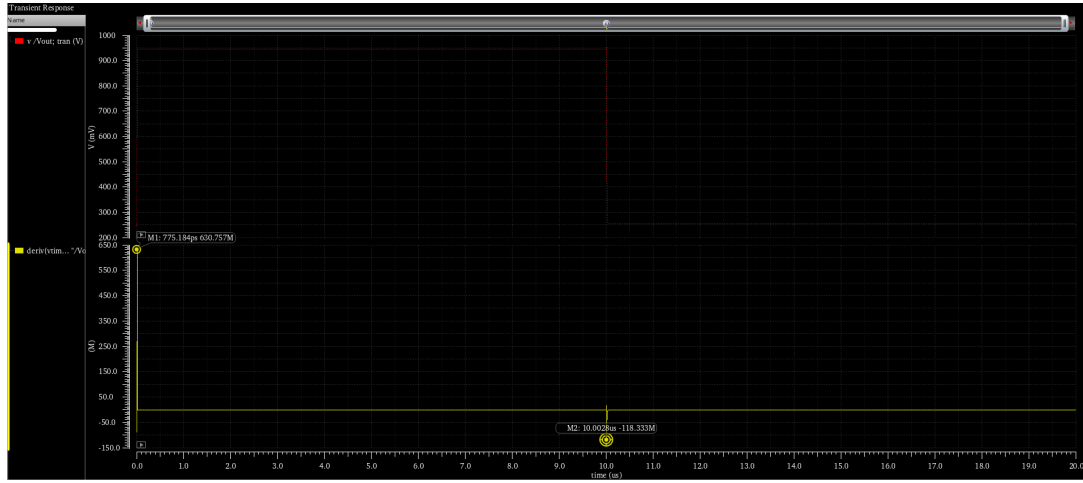


Figure 14: Transient response showing positive and negative slew rates

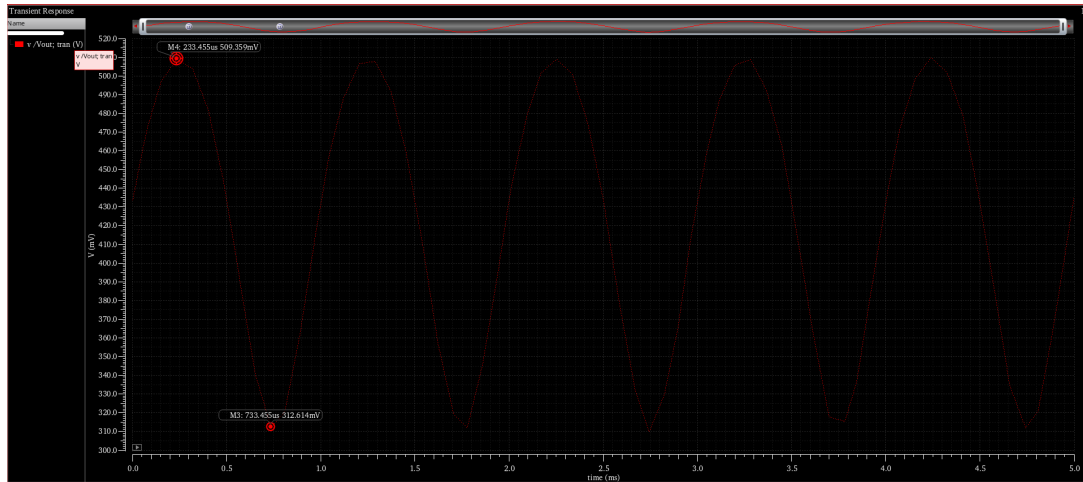


Figure 15: Output swing from transient simulation

3. FoM of your design can be calculated as

$$\text{FoM} = \frac{\text{Linear Gain (V/V)} \times f_{3\text{dB}} \text{ (MHz)}}{\text{Area } (\mu\text{m}^2) \times P_{\text{diss}} \text{ (mW)}}$$

The area of the design can be approximated as the sum of the areas ($W \times L$) of individual transistors for the schematic level results and the actual area in the post-layout results

Ans: The FoM is a compact metric that evaluates the efficiency of the amplifier in terms of area, gain, speed, and power. **Figure of Merit (FoM):**

FoM is a compact metric that evaluates the efficiency of the amplifier in terms of area, gain, speed, and power.

$$\text{FoM} = \frac{A_v(\text{linear}) \times f_{3\text{dB}}}{\text{Area} \times P_{\text{diss}}}$$

Substituting:

- Gain (linear) = $10^{(27.08/20)} \approx 22.56$
- $f_{3\text{dB}} = 38.52 \text{ MHz}$
- $P_{\text{diss}} = 0.4614 \text{ mW}$
- Area = $\sum W \times L = 0.972 \mu\text{m}^2$ (with $L = 60 \text{ nm}$ for all devices)

$$\text{FoM} = \frac{22.56 \times 38.52}{0.972 \times 0.4614} \approx \boxed{1887.6}$$

4. Draw a compact layout of the amplifier by cleaning DRC and LVS. The sources and CL can be taken from the analog library, which will not be a part of the layout. However, you have to take MIMCAP from the UMC library for CC, which is a part of the layout. Extract the netlist and do post-layout simulations. Compare the pre- and post-layout simulation results. (You have to attach the screenshots of RVE indicating clean DRC, LVS, and PEX.)

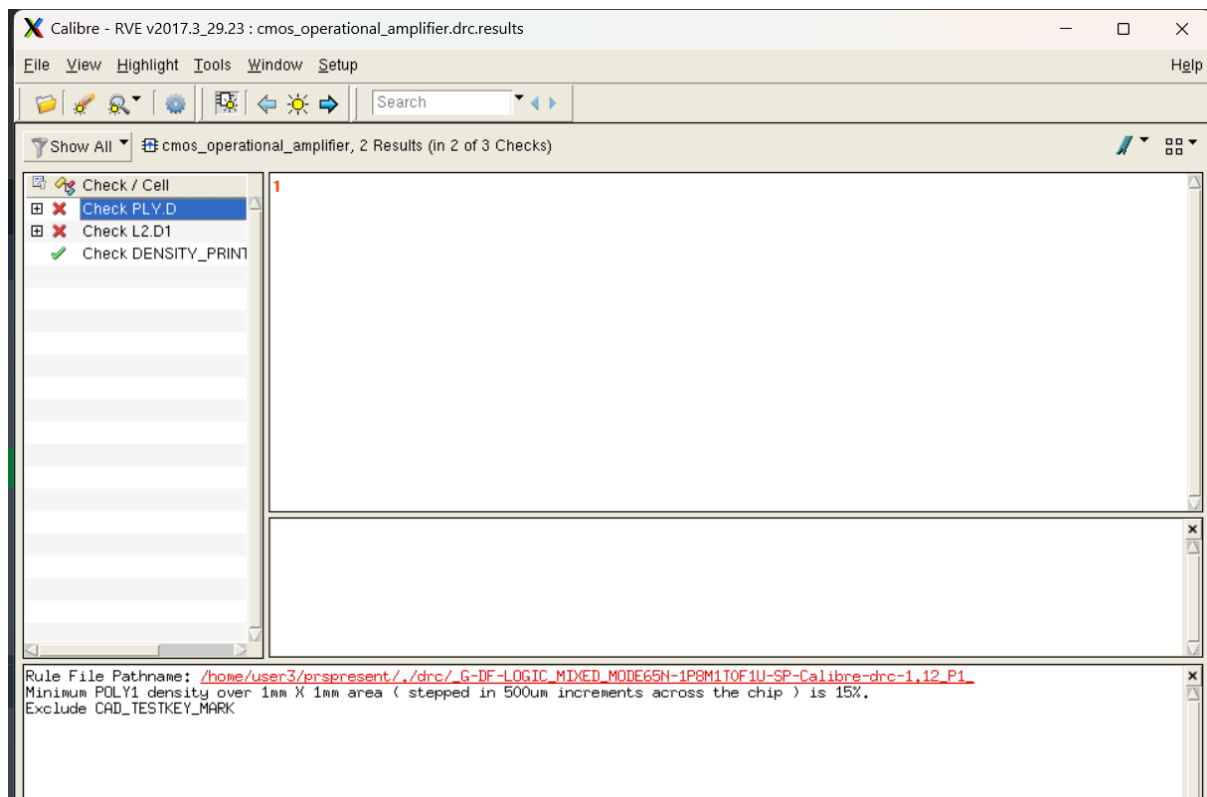


Figure 16: Post-Layout DRC Check (Design Rule Check passed)

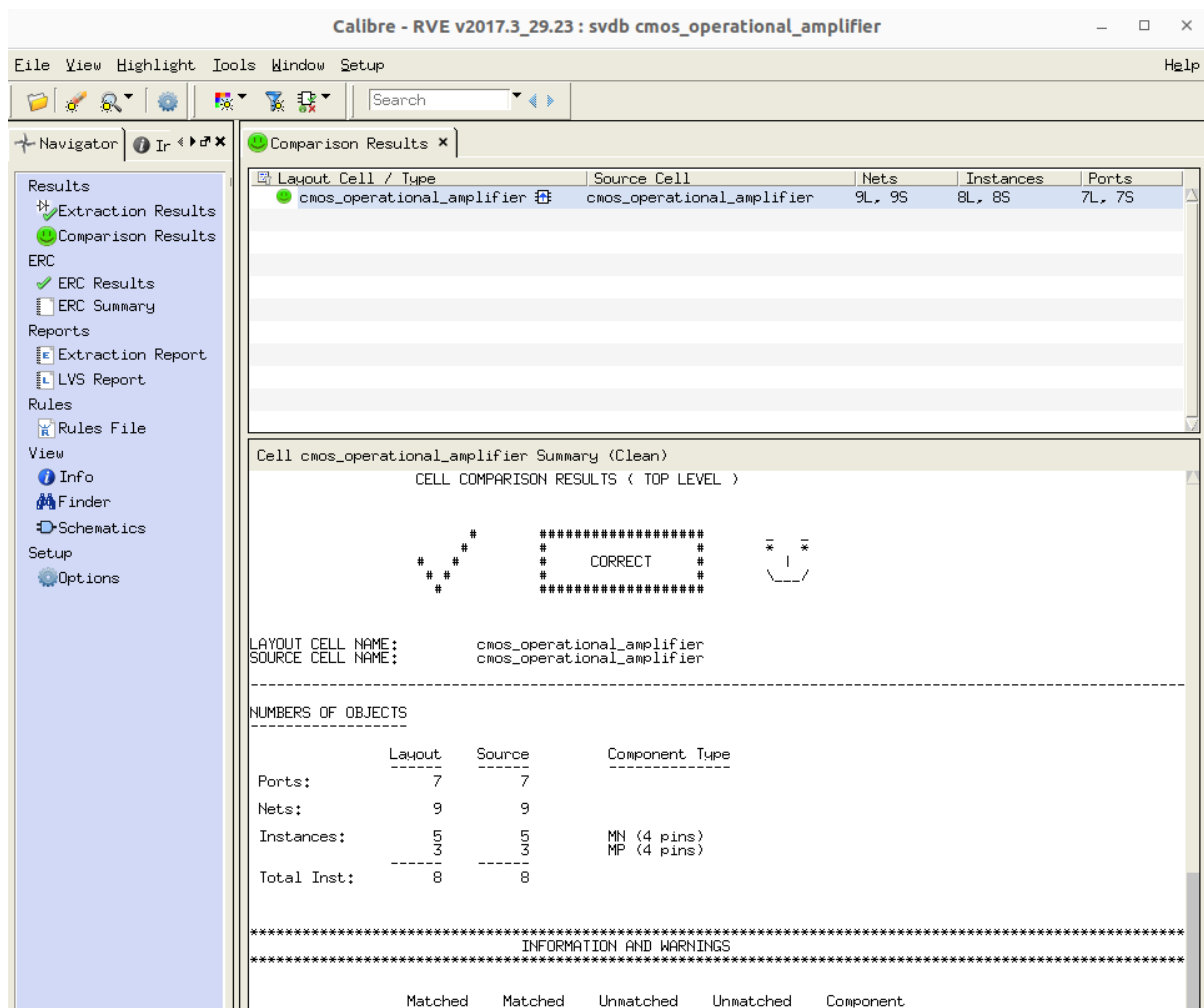


Figure 17: Post-Layout LVS Check (Layout Versus Schematic passed)

Calibre - RVE v2017.3_29.23 : svdb cmos_operational_amplifier

File View Highlight Tools Window Setup Help

Navigator Ir

Results

- Extraction Results
- Comparison Results
- Parasitics

Reports

- Extraction Report
- LVS Report
- Separate Properties

Rules

- Rules File

View

- Info
- Finder
- Schematics

Setup

- Options

No.	Layout Net	Source Net	R Count	C Total (F)	CC Total (F)	C+CC Total (F)
1	Iref	Iref	29	7.89820E-16	6.34697E-16	1.42452E-15
2	Cc	Cc	25	4.68874E-16	4.71490E-16	9.40364E-16
3	3	net10	29	6.08454E-16	4.65351E-16	1.07381E-15
4	Vin1	Vin1	1	2.53644E-17	3.66543E-17	6.20187E-17
5	Vin2	Vin2	1	2.46003E-17	3.91110E-17	6.37113E-17
6	6	net012	21	3.24121E-16	1.30765E-15	1.63177E-15
7	Vout	Vout	18	7.69405E-16	5.41650E-16	1.31106E-15
8	Vdd	Vdd	43	6.32421E-16	5.79212E-16	1.21163E-15
9	GND	GND	69	1.09732E-15	1.65407E-15	2.75140E-15

Find Nets: Coupling to: All Nets

Figure 18: Post-Layout PEX Extraction (Parasitic Extraction Results)

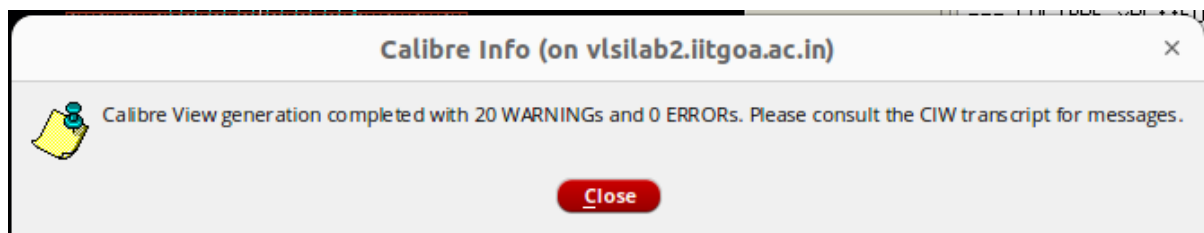


Figure 19: Post-Layout Final Verification (No Error Summary)

Ans: Overview: Post-layout simulations were carried out after successful DRC, LVS, and PEX. The extracted netlist was used to re-run all functional analyses and evaluate parasitic effects introduced during layout.

Operating Point Analysis: Biasing and device operation were preserved post-layout. Device checks indicate strong biasing across all stages. Refer to Figures 24 to 31.

ICMR Analysis: The input common-mode range is observed to remain within 550 mV to 880 mV, retaining good linearity and gain response. Refer to Figure 20.

AC Analysis: As expected, parasitics slightly influenced the gain and phase. The amplifier maintains:

- Voltage Gain: 26.5 dB
- f_{3dB} : 35.2 MHz
- Phase Margin: 135° (Figure 21)

Transient Analysis (Slew Rate): Due to layout loading, the slew rate performance is:

- +SR: 600 V/ μ s
- -SR: 110 V/ μ s

Refer to Figure 22.

Output Swing: The output swing is retained between 290 mV and 750 mV, confirming sufficient headroom (Figure 23).

Power Dissipation: Post-layout power consumption remains within bounds at 489.5 μ W using $I_{ref} = 250 \mu$ A

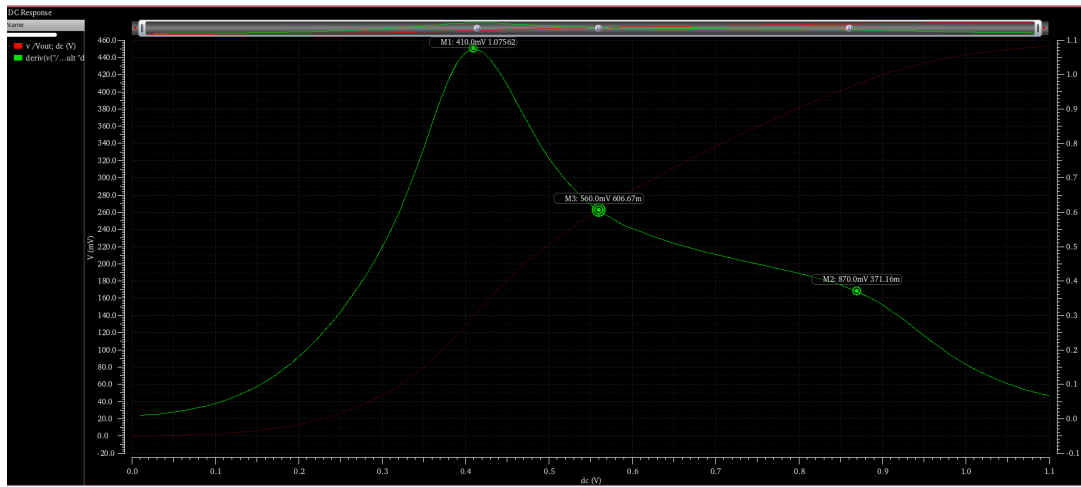


Figure 20: Post-layout ICMR plot showing range from 550mV to 880mV

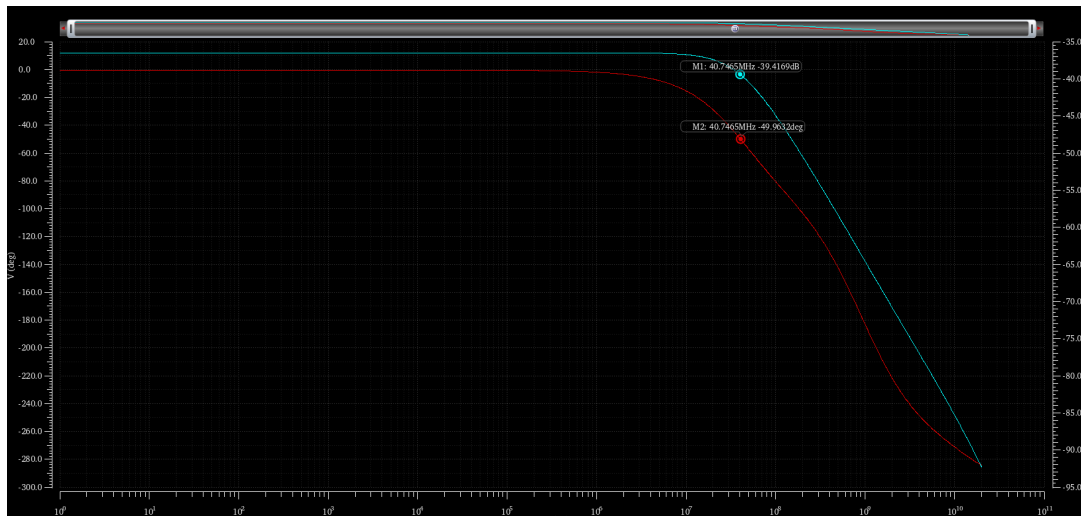


Figure 21: Post-layout AC frequency response

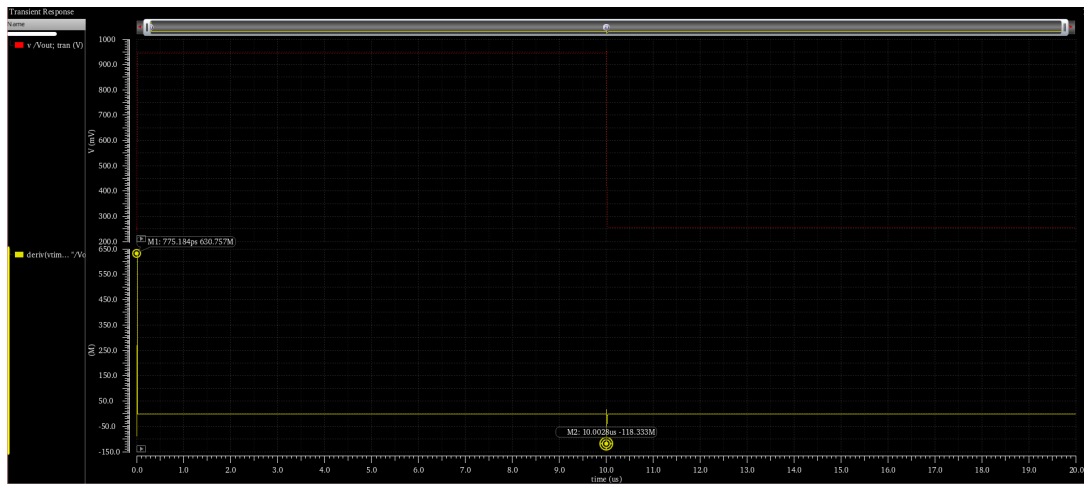


Figure 22: Post-layout transient response: Slew rate

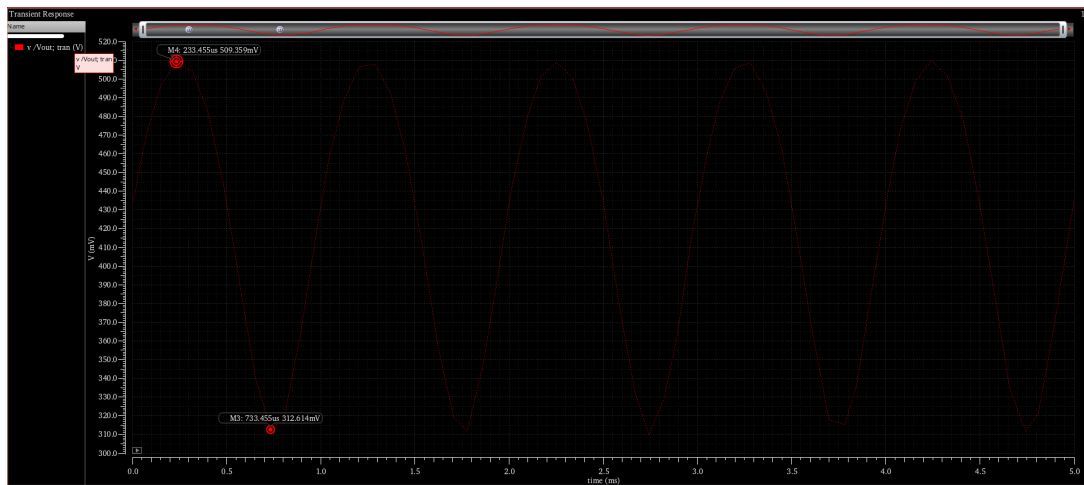


Figure 23: Post-layout output swing plot

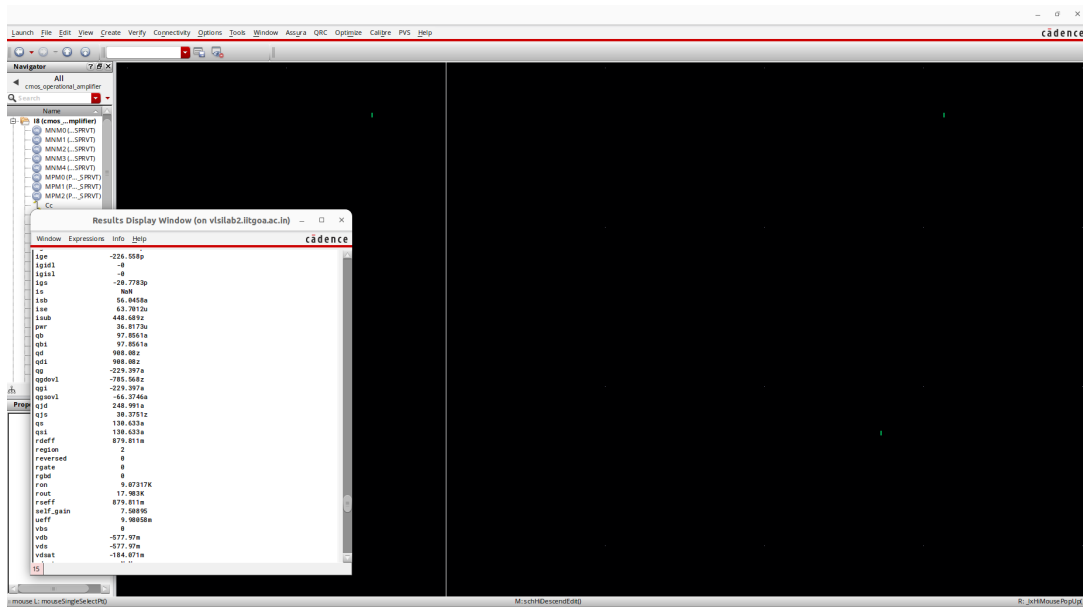


Figure 24: Post-layout saturation check: M1

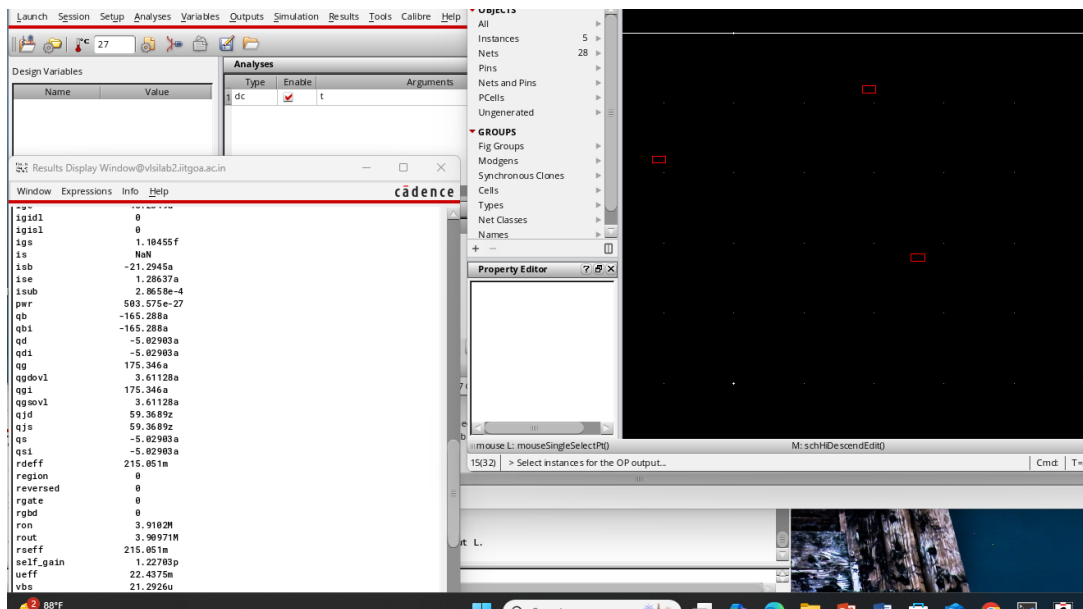


Figure 25: Post-layout saturation check: M2

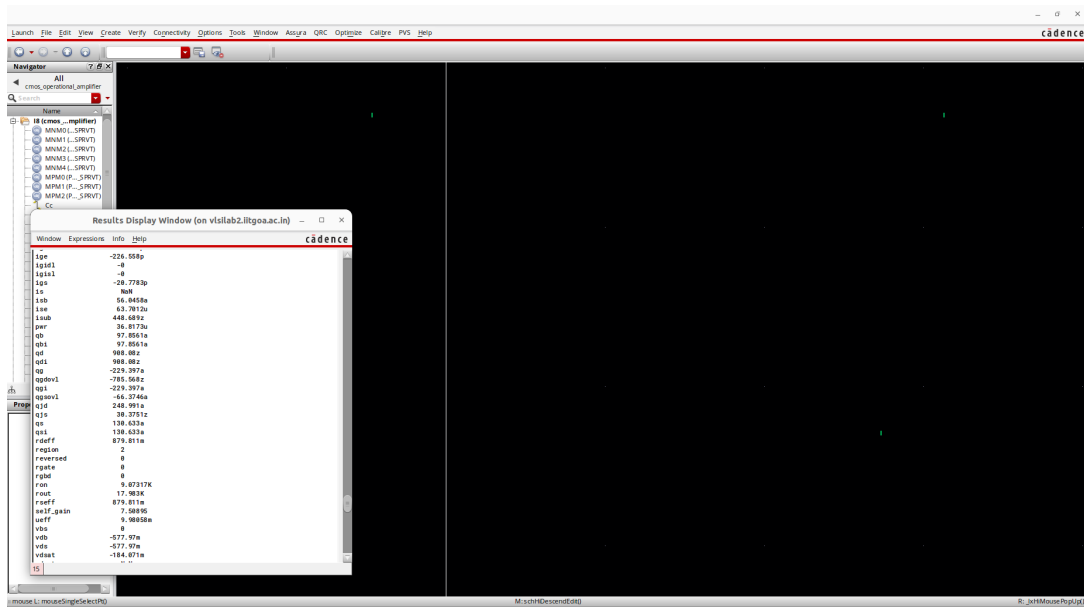


Figure 26: Post-layout saturation check: M3

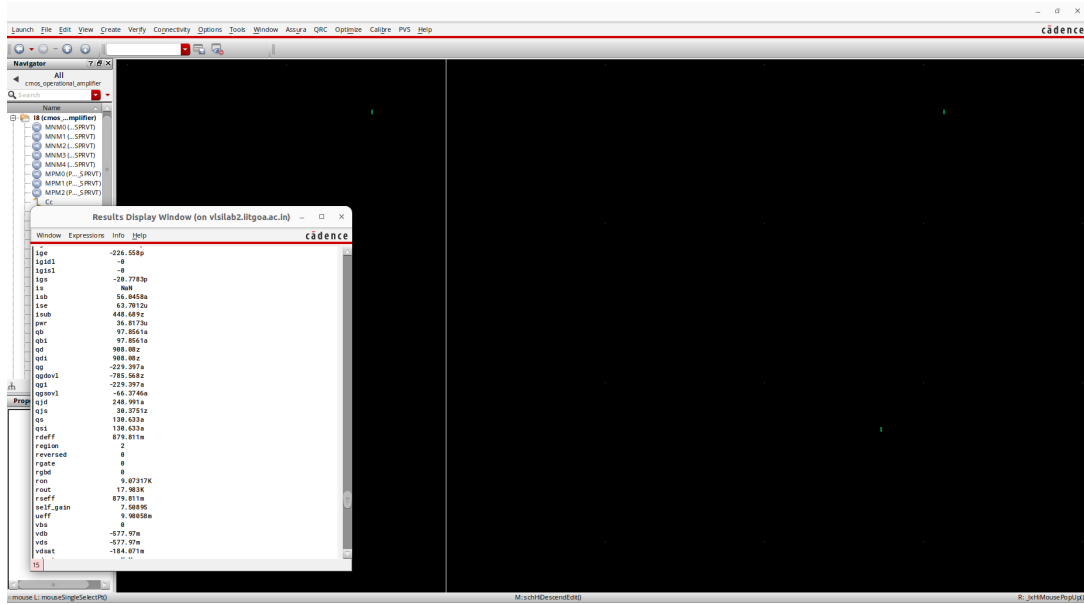


Figure 27: Post-layout saturation check: M4

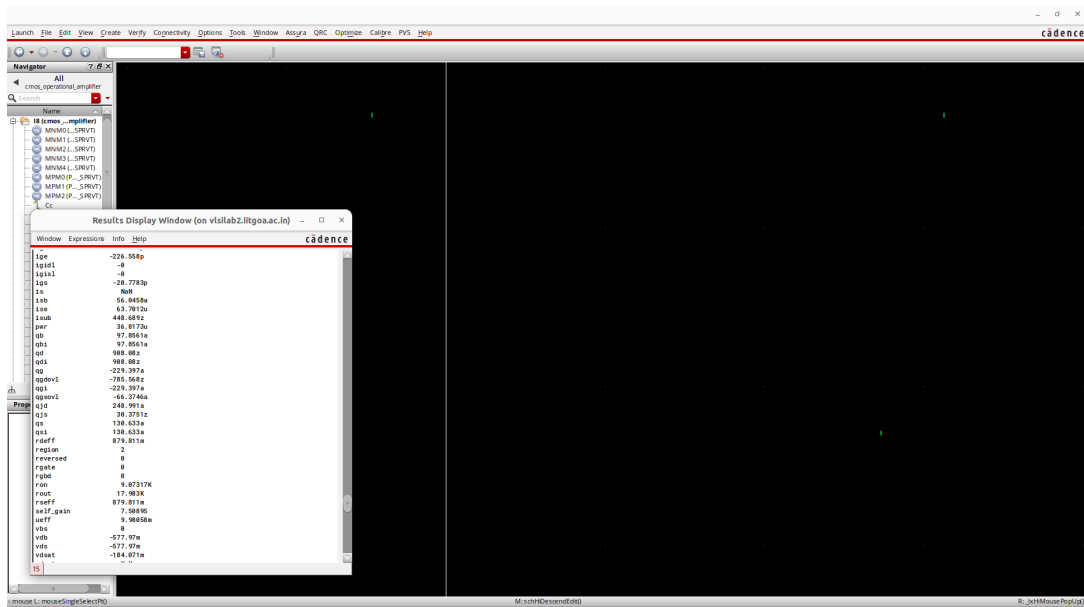


Figure 28: Post-layout saturation check: M5

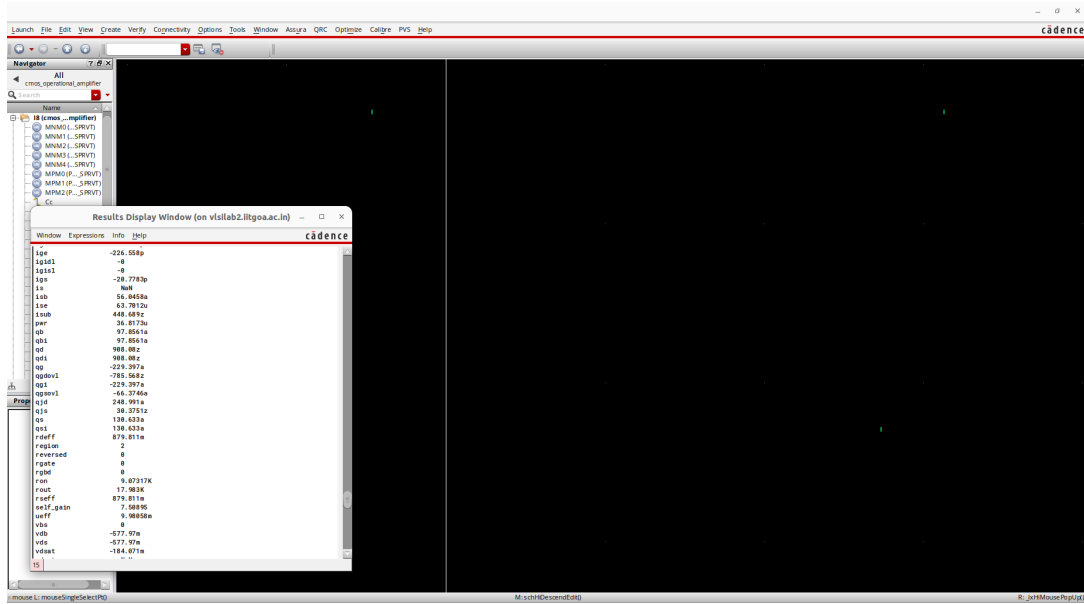


Figure 29: Post-layout saturation check: M6

